Grade 10 MYP Physics Criteria B & C Summative Assessment (Mr. Pathan)

IB Middle Years
Physics

Grade: Grade 10 MYP

Criteria: B 17 C 21

Total: 38 points

Duration: 50 mins

Close time: Oct 09, 11:32 am (School TZ)

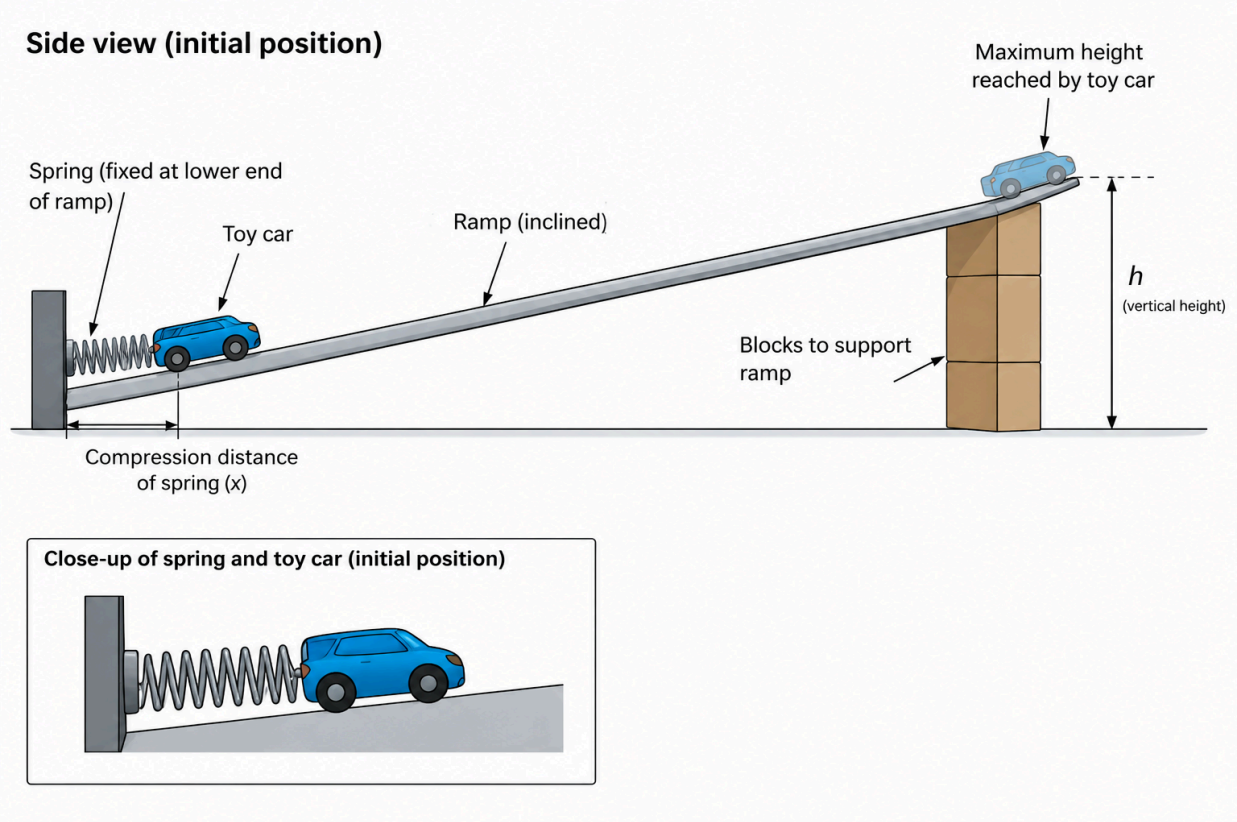
Experiment on Conservation of Energy

A group of students wants to investigate the transfer of energy from elastic potential energy to gravitational potential energy.

The students have the following equipment:

- a toy car
- a spring
- an inclined ramp
- a protractor (not shown in the diagram)
- blocks to support the ramp

The spring is fixed at the lower end of the ramp. The toy car is placed against the spring and the spring is compressed by a measured distance. When released, the spring pushes the car up the ramp. The car eventually stops at its maximum height.



Q 1.1 **Design** an investigation to determine the relationship between the compression of the spring and the maximum gravitational potential energy gained by the toy car. B 16

Your investigation design should include:

- research question **[1]**
- independent and dependent variables **[1]**

- two relevant controlled variables and justification for controlling these variables **[4]**
- additional two equipment required to carry out the investigation **[2]**
- a clear and detailed procedure **[5]**
- a description of the data to be collected for the investigation to be reliable. **[3]**

Markscheme

	1 mark	2 marks	3 marks	4 marks	5 marks
RQ	States a question involving **compression distance of spring** and **maximum gravitational energy attained by the car**.				
IV & DV	Identifies the independent (compression distance) **AND** dependent variable (maximum height).				
CVs	Identifies **one relevant controlled variable** without justification.	Identifies **two relevant controlled variables** without			

		justificatio n.	Identifies one relevant controlled variable and gives a relevant justificatio n for **one** controlled variable.	Identifies two relevant controlled variables and gives relevant justificatio ns for **both** controlled variables.	
Eq	Identifies suitable additional equipment needed to measure **height**, e.g., ruler	Identifies suitable additional equipment needed to **height**, e.g., ruler, **AND** mass, e.g., digital weighing scale OR balance			

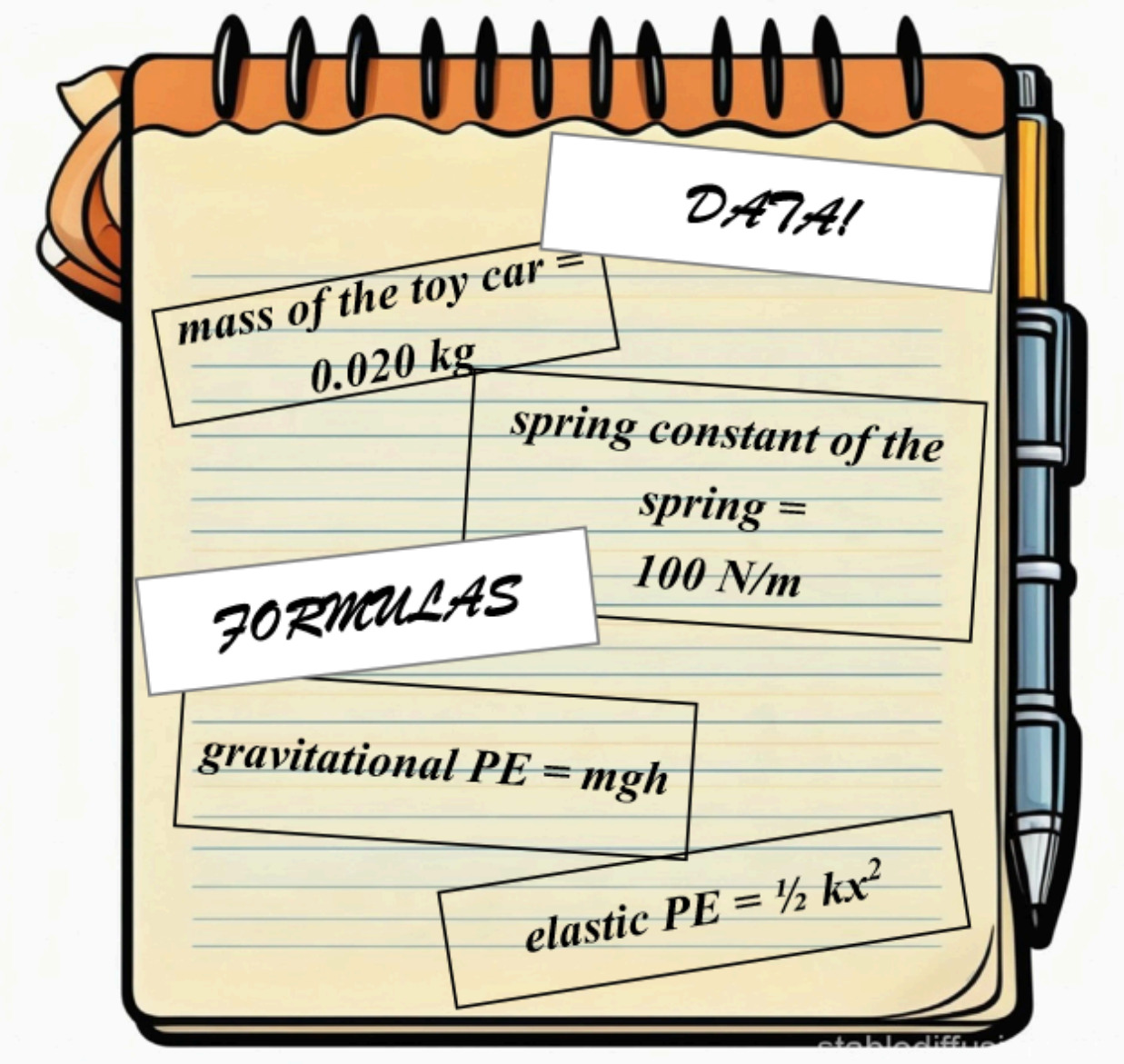
P	Describes how the spring is compressed consistently at the first initial compression distance, while controlling CVs.	Describes how to create and **measure different compression distance**.	Describes how the car is **released consistently** and how its maximum height is measured (from a reference point).	Describes repeating the procedure for **different compression distances** while controlling other variables.	Gives a **complete, logical procedure**, including conducting multiple trials **AND** calculating the average speed.
D	States that **at least 5 different compression distance** should be tested and measurements repeated.	States that **at least 5 different compression distance** should be tested and measurements repeated **AND** specifies a suitable range of compression and **at least three trials per height**.	States that **at least 5 different compression distance** should be tested and measurements repeated **AND** specifies a suitable range of compression and **at least three trials per height.t**, with a mean/average calculated.		

Another group carried out the investigation to determine the elastic potential energy stored in the spring and the gravitational potential energy stored in the toy car at the end of the ramp.

Their results are shown below.

Compression of spring / m	Maximum vertical height gained / m
0.04	0.031
0.06	0.071
0.08	0.122
0.10	0.199
0.12	0.296

The following data are also obtained by the group, together with formulas for the elastic potential energy and gravitational potential energy:



The group formulates a hypothesis as:

"The elastic potential energy stored in the spring is not equal to the gravitational potential energy achieved by the toy car when it reaches the maximum height because some of the mechanical energy will be lost as it moves up the ramp."

Q 1.2 **State** a conclusion about whether the results support the group's hypothesis. **Use** evidence from the data obtained by the group to justify your conclusion. C 2

Markscheme

1 mark — The results support the hypothesis

1 mark — Evidence for the energy relationship: Uses data to show that GPE on top is not equal to EPE of the spring.
For example, at 0.04 m of compression distance, EPE = 0.08 J while GPE = 0.04 J.
Therefore, the claim that the GPE is equal to the KE is not supported.
Do not accept the first mark without a sample working.

Q 1.3 A member of the group suggested that there are a number of sources of systematic error in the experimental setup that could explain their results. C 2

Identify one major source of systematic error from the experimental setup. **Explain** its impact on the results of the experiment.

Markscheme

1 mark — **The cart did not start from the very bottom of the ramp**
1 mark — meaning its initial gravitational potential energy was not zero/was not accounted for *owtte*.

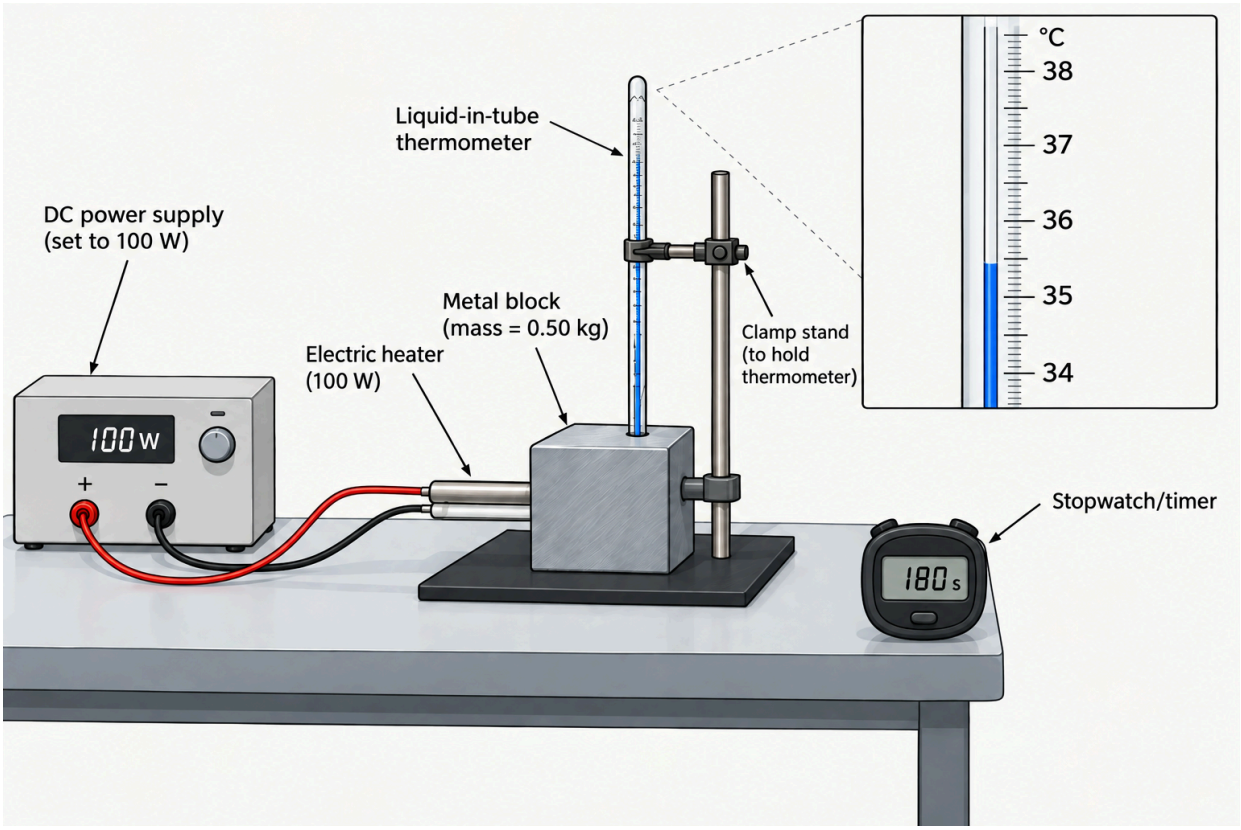
Award max 1 mark for ANY logical systematic error from the setup.
For example,
The ramp was not perfectly smooth, so friction acted on the cart as it travelled up the ramp, causing some energy to be transferred to the surroundings.

Experiment on Specific Heat Capacity

A group of students investigates the specific heat capacity of a metal block. They use an electric heater to transfer energy to the block and measure its temperature at regular time intervals.

The mass of the metal block is **0.50 kg**. The heater has a power of **100 W**.

The experimental setup is shown with the metal block is uninsulated as it lays on the base of the black metal retort stand.



Q 2.1 **State** one safety consideration that should be followed when carrying out this investigation. B 1

Markscheme

Any valid safety consideration. Examples include, but not limited to:

- Use heat-resistant gloves or tongs when handling the heated metal block.
- Do not touch the electric heater while it is switched on.
- Allow the metal block to cool before handling it.
- Keep electrical connections away from water or other liquids.
- Switch off the power supply before adjusting or removing the apparatus.
- Handle the thermometer carefully to avoid breaking it.

Q 2.2 The temperature is measured every 1 minute. C 4

The temperature at t = 180 s is shown in the diagram.

Measure the temperature at t = 180s. **Complete** the table, including the column heading and the missing time when the temperature is 43.9 degree Celsius.

0	22.0
60	26.5
120	31.0
180	
240	39.7
	43.9
360	48.0

Markscheme

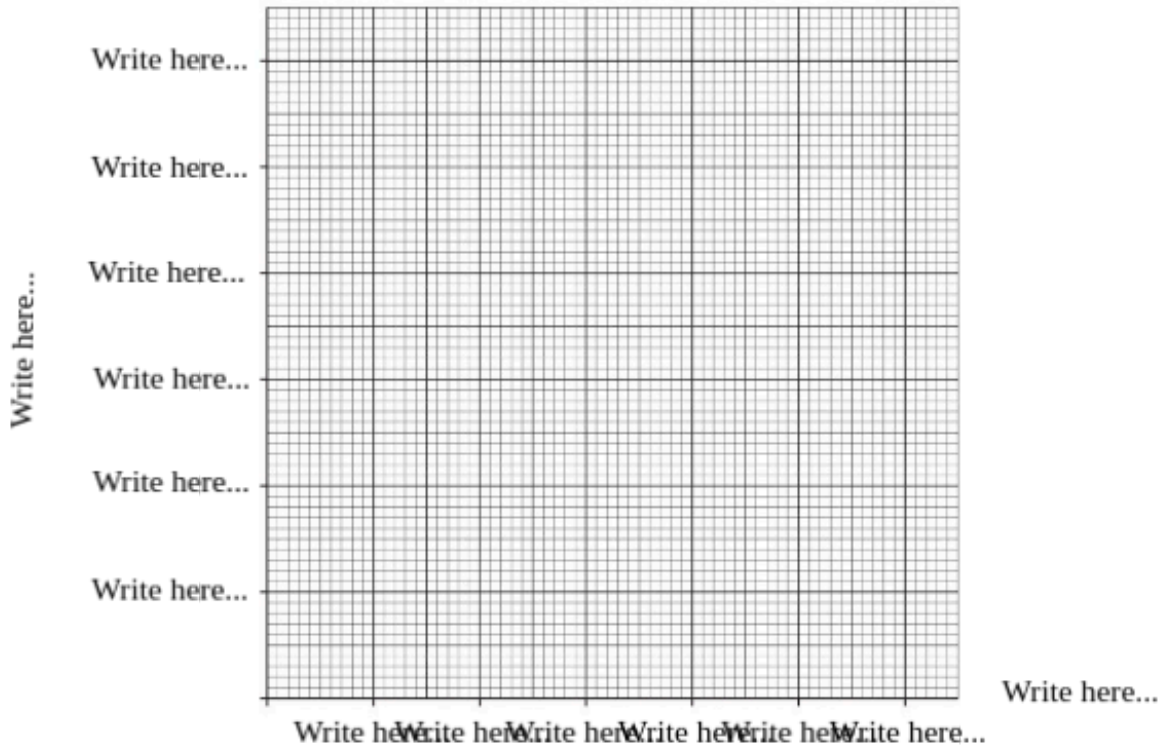
1 mark — time in first column heading AND temperature in second column heading

1 mark — both units correct; second for time AND degree Celsius for temperature

1 mark — 35.4 (temperature at t = 180 s)

1 mark — 300 (time when T = 43.9 degree Celsius)

Q 2.3 **Plot** a line graph to show the relationship between time of heating and temperature of the metal block. Start your graph at (0,0). C 4



Markscheme

- **1 mark** — Time of heating end plotted on the x-axis and temperature on the y-axis, both with units
- **1 mark** — Appropriate, even scales are used and occupy a reasonable proportion of the graph, i.e, at least half of the grid
- **1 mark** — All six points are plotted accurately (plus/minus 1 small square)
- **1 mark** — Suitable line of best fit is drawn.

Q 2.4 **Show that** the rate of change in temperature $\frac{\Delta T}{\Delta t}$ is about 0.07 degree Celsius per second. C 2

Markscheme

1 mark — states that rate of change in temperature $\frac{\Delta T}{\Delta t}$ = gradient of the line

1 mark — shows calculation of gradient of line; two points must be at least half of the line apart.

Q 2.5 The relationship between the power P supplied to a substance, its mass m, its specific heat capacity c, and the rate of change in temperature $\frac{\Delta T}{\Delta t}$ is given by: C 2

$$P = m \times c \times \frac{\Delta T}{\Delta t}$$

where ***P*** is power, ***m*** is mass, and ***c*** is specific heat capacity.

Calculate the specific heat capacity of the metal using this equation. Include its unit.

Markscheme

1 mark — Correctly rearranges the equation: **c = P ÷ (m × rate of change in temperature)**
1 mark — Correct answer: **c = 2857 J kg^{-1} °C^{-1} ≈ 2900 J kg^{-1} °C^{-1}** or **2.9 × 10^{3} J kg^{-1} °C^{-1}**

Q 2.6 The student suggested that the calculated specific heat capacity of the metal block in Q2.5 is less than the actual specific heat capacity of the metal block. C 2

Explain this suggestion.

Markscheme

1 mark — States that the metal block is uninsulated OR that some of the energy supplied by the heater is transferred to the surroundings rather than the metal block
1 mark — Explains that this makes the measured rate of temperature increase **smaller**, so using **P = m × c × rate of change in temperature** would give a calculated specific heat capacity **greater than**, not less than, the actual value.

Q 2.7 The student wants to extend the investigation a with a new pair of independent variables to determine the specific heat capacity of a metal block. C 3

The new independent variable is the electric power of the source, and the new dependent variable is temperature of the heating, while controlling the duration of heating

Formulate a scientific hypothesis for this new set of variables.

IF:	THEN:	BECAUSE:

Markscheme

IF-THEN
1 mark —
IF: States that increasing the **electric power** increases the temperature of the metal block
THEN: Predicts that the **temperature of the metal block will be higher** after the same fixed duration of heating.

BECAUSE:

2 marks —

CORRECT BECAUSE Statement: Explains that greater power transfers **more energy per unit time** to the metal block, resulting in a greater temperature increase; award only 1 mark if logical BECAUSE statement but not stated with scientific terminologies.

Max 1 mark —

For an incorrect BECAUSE Statement but logically and scientifically related justification/explanation.